SZCZUREK, SANTINI, FIACCHI - HM3

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1. *Use the rejection algorithm to generate samples from the N(0,1) density. Try the following proposals g(x):*

(a) T-Student distribution with 2 degrees of freedom (T2) (you can use functions rt and dt);
(b) Logistic distribution (use rlogis and dlogis functions)

For both proposals, plot these two densities and find the optimal value of the constant c. Plot in the same graph the corresponding envelope function. Generate n = 10000 random variates by using an implementation of the rejection algorithm. Compute the acceptance rate and compare with the true probability of acceptance.

Compare the above results with the corresponding output of the function AR.Sim of the package AR.

Which is the best proposal? Why?

```
SET.SEED(123)
N <- 10000
#TARGET FUNCTION (N(0,1))
F <- FUNCTION(X) DNORM(X)

#PROPOSALS

#T-STUDENT STANDARD (DF=2)
G1 <- FUNCTION(X) DT(X, DF = 2)
R_G1 <- FUNCTION(N) RT(N, DF = 2) #GENERATOR OF T-STUDENT

#LOGISTIC
G2 <- FUNCTION(X) DLOGIS(X)
R_G2 <- FUNCTION(N) RLOGIS(N)

#DEFINE X
X <- SEQ(-10, 10, LENGTH.OUT = 10000)

#RATIO
FG1 <- FUNCTION(X) {F(X)/G1(X)}
```

```r
#OPTIMAL C
SET.SEED(123)

MAX1 <- OPTIMIZE(FG1,INTERVAL=C(-10,10),MAXIMUM=TRUE)

C1 <- MAX1$OBJECTIVE
C1

## [1] 1.257317

#TRUE ACCEPTANCE RATE
1/C1

## [1] 0.7953445

#ENVELOPE FUNCTION
CG1 <- FUNCTION(X){C1*G1(X)}

#LOGISTIC
FG2 <- FUNCTION(X) {F(X)/G2(X)}

#OPTIMAL C
SET.SEED(123)

MAX2 <- OPTIMIZE(FG2,INTERVAL=C(-10,10),MAXIMUM=TRUE)

C2 <- MAX2$OBJECTIVE
C2

## [1] 1.595769

#TRUE ACCEPTANCE RATE
1/C2

## [1] 0.6266571

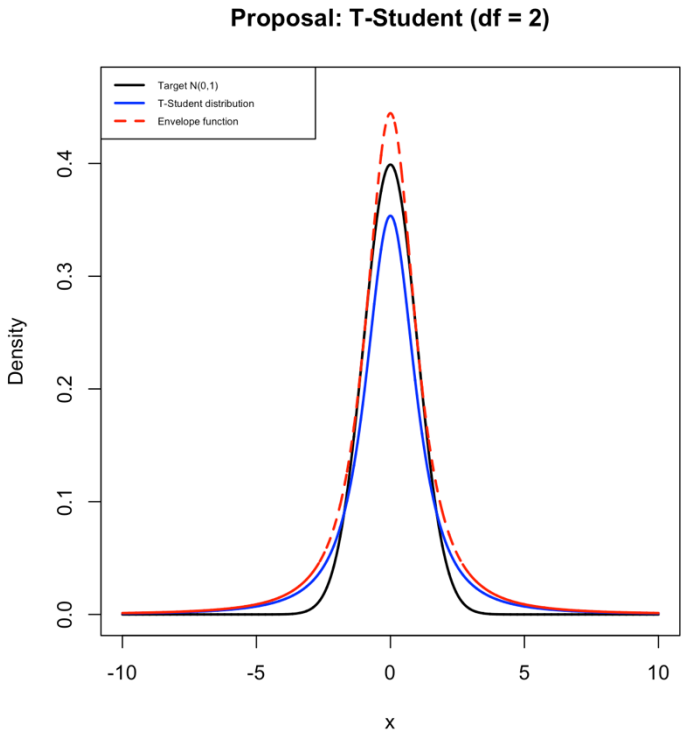
#ENVELOPE FUNCTION
CG2 <- FUNCTION(X){C2*G2(X)}

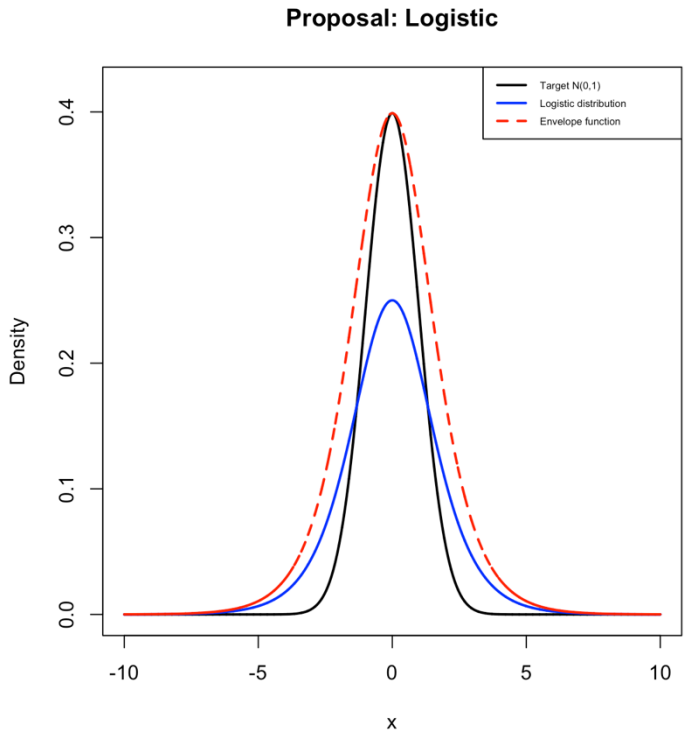
#PLOT
PAR(MFROW = C(1,2))

#T-STUDENT
PLOT(X, F(X), TYPE = "L", COL = "BLACK", LWD = 2, YLIM = C(0, MAX(CG1(0), F(
0)) * 1.05),
     MAIN = "PROPOSAL: T-STUDENT (DF = 2)", XLAB = "X", YLAB = "DENSITY")
LINES(X, G1(X), COL = "BLUE", LWD = 2)
LINES(X, CG1(X), COL = "RED", LWD = 2, LTY = 2)
LEGEND("TOPLEFT", LEGEND = C("TARGET N(0,1)", "T-STUDENT DISTRIBUTION", "ENV
```

```
ELOPE FUNCTION"),
       COL = C("BLACK","BLUE","RED"), LWD=2, LTY=C(1,1,2), CEX = 0.5)

#LOGISTIC
PLOT(X, F(X), TYPE = "L", COL = "BLACK", LWD = 2, YLIM = C(0, MAX(CG2(0), F(
0)) * 1.05),
     MAIN = "PROPOSAL: LOGISTIC", XLAB = "X", YLAB = "DENSITY")
LINES(X, G2(X), COL = "BLUE", LWD = 2)
LINES(X, CG2(X), COL = "RED", LWD = 2, LTY = 2)
LEGEND("TOPRIGHT", LEGEND = C("TARGET N(0,1)", "LOGISTIC DISTRIBUTION", "ENV
ELOPE FUNCTION"),
       COL = C("BLACK","BLUE","RED"), LWD=2, LTY=C(1,1,2), CEX = 0.5)
```





From these theoretical calculations, we can already state that the proposal using the t distribution is preferable, since it provides a smaller constant c1 compared to c2 (which means a better envelope function) and, consequently, a higher acceptance rate. This implies that the t-Student proposal generates more accepted samples and fewer rejections, making it more efficient than the other one. Moreover, the t-Student distribution has heavier tails than the normal, which allows it to better cover the target N(0,1), especially in the tails. The logistic distribution, instead, has lighter tails, which means it needs to be scaled up more (indeed, it uses a higher c2) to fully enclose the target N(0,1).

```
###### --- REJECTION SAMPLING ALGORITHM --- ######
SET.SEED(123)

REJECTSAMP <-FUNCTION(N, C, F, G, R_G){
  X <- NUMERIC(N); K <- 0; J <- 0
```

```
  WHILE (K < N){
    J <- J + 1
    Y <- R_G(1)
    U <- RUNIF(1)
    IF(U<=F(Y)/(C*G(Y))){
      K <- K + 1
      X[K] <- Y
    }
  }
  OUT <- LIST(X=X,J=J)
  RETURN(OUT)
}

#T-STUDENT
RJ1 <- REJECTSAMP(N=10000,C=C1,F=F,G=FUNCTION(X) DT(X, DF = 2),R_G =FUNCTION
(N) RT(N, DF = 2))


#TRUE ACCEPTANCE RATE
1/C1
```

```
## [1] 0.7953445
```

```
#ACCEPTANDE RATE
N/RJ1$J
```

```
## [1] 0.7971939
```

```
#LOGISTIC
RJ2 <- REJECTSAMP(N=10000,C=C2,F=F,G=FUNCTION(X) DLOGIS(X),R_G =FUNCTION(N)
RLOGIS(N))


#TRUE ACCEPTANCE RATE
1/C2
```

```
## [1] 0.6266571
```

```
#ACCEPTANCE RATE
N/RJ2$J
```

```
## [1] 0.6298419
```

```
#TABLE FOR A CLEAR VISUALIZATION
TAB <- DATA.FRAME(
  T_STUDENT_DISTRIBUTION = C(1/C1, N/RJ1$J),
  LOGISTIC_DISTRIBUTION  = C(1/C2, N/RJ2$J),
  ROW.NAMES = C("TRUE_PROBABILITY_OF_ACCEPTANCE", "ACCEPTANCE_RATE")
)

TAB
```

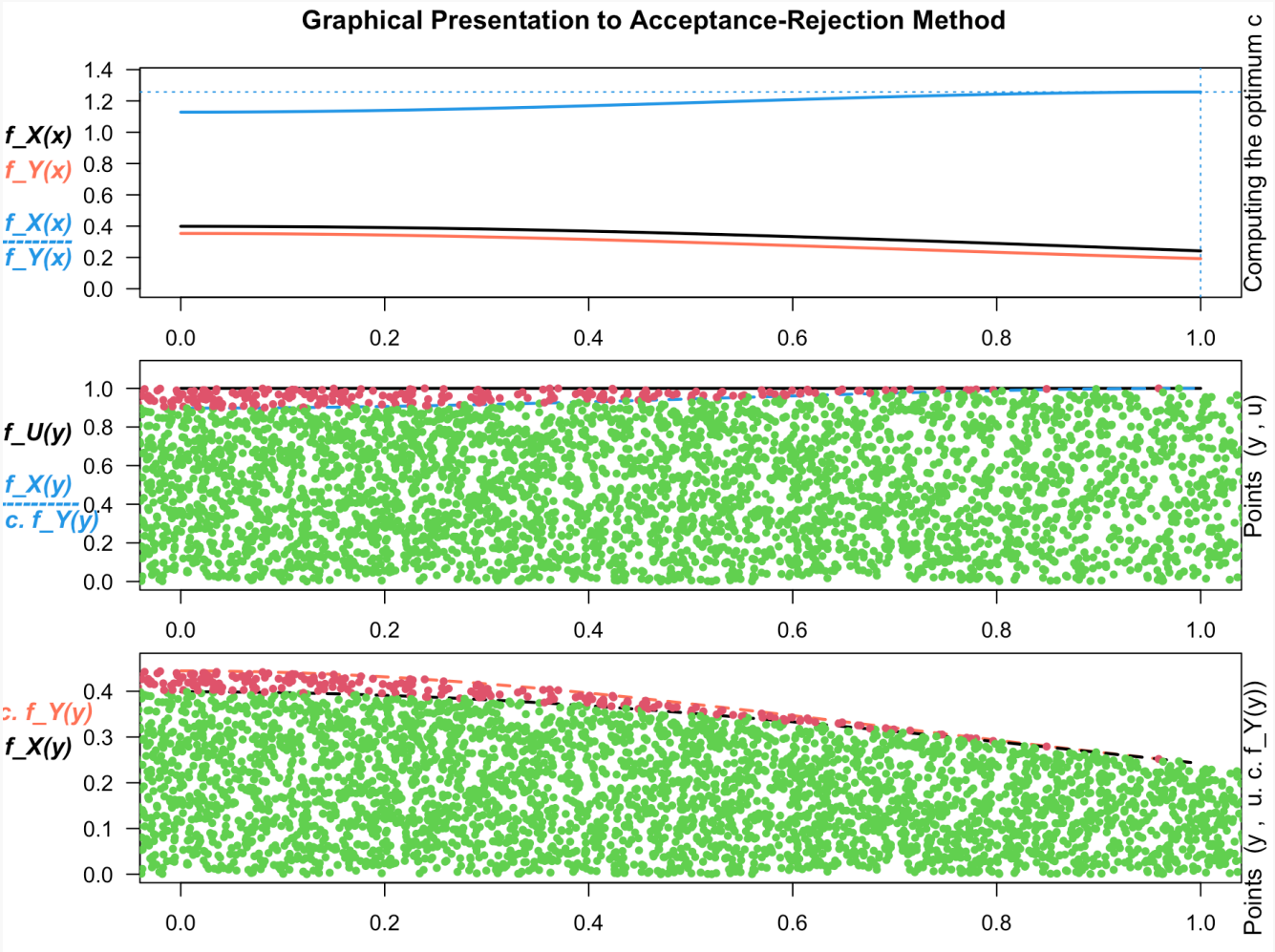
```
##                                 T_STUDENT_DISTRIBUTION LOGISTIC_DISTRIBUTI
ON
## TRUE_PROBABILITY_OF_ACCEPTANCE              0.7953445             0.62665
71
## ACCEPTANCE_RATE                             0.7971939             0.62984
19
```

```r
#PACKAGE AR.SIM
LIBRARY(AR)
SET.SEED(123)

#T STUDENT
SIM_1 <- AR.SIM(N = N, F_X = F, Y.DIST = "T", Y.DIST.PAR = 2)
```

```
## OPTIMAL C = 1.257

## THE NUMBERS OF REJECTIONS = 2624
## RATIO OF REJECTIONS = 0.208
## RATIO OF ACCEPTANCE = 0.792
```

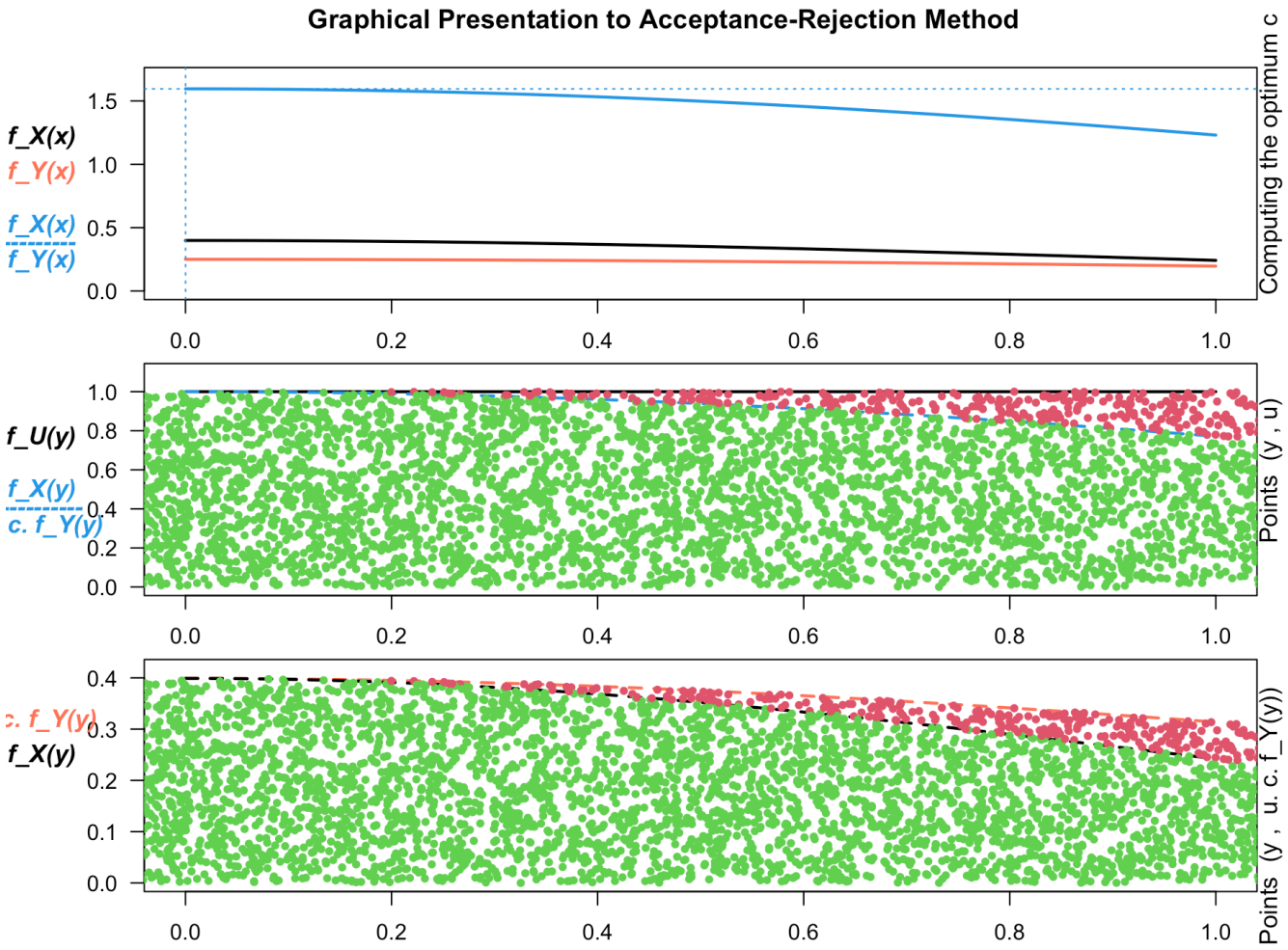


```r
#LOGISTIC
SET.SEED(123)

SIM_2 <- AR.SIM(N = N, F_X = F, Y.DIST = "LOGIS", Y.DIST.PAR = C(LOCATION =
0, SCALE = 1))

## OPTIMAL C = 1.596

## THE NUMBERS OF REJECTIONS = 5833
## RATIO OF REJECTIONS = 0.368
## RATIO OF ACCEPTANCE = 0.632
```



It was previously hypothesized that the proposal using the t distribution would be better than the logistic one, since its envelope function is more closely aligned with the target distribution. This results in a higher acceptance probability compared to the other proposal.

Afterwards, both by implementing the rejection algorithm manually and by using the AR.Sim function, our initial hypothesis can be confirmed. By comparing the results obtained with the AR package, we can observe that the number of rejected samples is much lower when using the t-Student proposal compared to the logistic

distribution, and consequently, the number of accepted samples is higher for the former. These results are also supported by the graphical analysis, since in the first case the red area (representing rejections) is much smaller than in the other one.

2. *Explain how to generate a random variable with the density*

$$f(x) = 2.5x\sqrt{x},\ 0 < x < 1$$

if your random number generator produces a Uniform random variable U~Unif(0,1).

```
# WE GENERATE N SAMPLES FROM THE DENSITY F(X) = 2.5*X*SQRT(X), 0 < X < 1

U <- RUNIF(N)
X <- U^(2/5)
```

In this exercise we want to simulate a random variable X whose probability density (pdf) is:

$$f(x) = 2.5x\sqrt{x} = 2.5x^{3/2},\ 0<x<1.$$

That means we want to produce random values $X1, X2, \ldots, Xn$ that follow this distribution. The problem is that our random number generator produces only uniform random numbers between 0 and 1. So we need to transform this uniform variable U into a new variable X that has the desired distribution.

The main idea is to apply the inverse transformation method but to compute it, we need the CDF (the integral of the density). Once we have obtained the cumulative, we can find the inverse of F(x).

Setting U=F(x)=x^5/2", we solve for x, and we have: x=U^2/5 (inverse).

This means that if U is uniformly distributed on (0,1), then X=U^2/5 follows the target distribution with pdf f(x)=2.5x^3/2.

3. *Use Monte Carlo integration to estimate the following integrals. Compare with the exact answer, if known, or with the deterministic numerical integration. Monitor the convergence to the true value with the 95% confidence bounds (with an appropriate graph);*

(a) $\int_1^3 x^2 dx$

(b) $\int_0^\pi \sin(x) dx$

(c) $\int_0^2 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$

(d) $\int_0^{1/20} \frac{\exp(-\frac{1}{2x^2})}{x^2\sqrt{2\pi}}\, dx$

(e) $\int\int_R xy(x^2+y^2)\, dxdy \quad R = \{(x,y) : 0 \le x \le 1,\ 0 \le y \le 1\}$

(f) $\int\int_R x\sin(xy)\, dxdy \quad R = \{(x,y) : 0 \le x \le 1,\ \pi \le y \le 2\pi\}$

(g) $\int\int_R x^2 y \exp(xy)\, dxdy \quad R = \{(x,y) : 0 \le x \le 1,\ 0 \le y \le 2\}$

(h) $\int\int_R x\cos(x^2+y)\, dxdy \quad R = \{(x,y) : 0 \le x \le \pi,\ -\sqrt{\pi} \le y \le 0\}$

```
FUNCTION_A<-FUNCTION(X)(X^2)
FUNCTION_B<-FUNCTION(X)(SIN(X))
FUNCTION_C<-FUNCTION(X) ((1 / SQRT(2*PI)) * EXP(-X^2 / 2))
FUNCTION_D<-FUNCTION(X) (EXP(-1/(2*X^2)) / (X^2 * SQRT(2*PI)))
FUNCTION_E<-FUNCTION(X,Y) (X*Y*(X^2 + Y^2))
FUNCTION_F<-FUNCTION(X,Y) (X*SIN(X*Y))
FUNCTION_G<-FUNCTION(X,Y) (X^2*Y*EXP(X*Y))
FUNCTION_H<-FUNCTION(X,Y) (X*COS(X^2 + Y))

#MC FOR SINGLE INTEGRALS
#MINY AND MAXY ARE GRAPHICAL PARAMETERS
INTEGRATION1 <- FUNCTION(G, LOWLIM, UPLIM, MINY, MAXY) {
  SET.SEED(12)
  F <- FUNCTION(X) DUNIF(X,LOWLIM,UPLIM)
  X <- RUNIF(10^5,MIN=LOWLIM,MAX=UPLIM)
  HAT.THETA <- MEAN(G(X)/F(X))

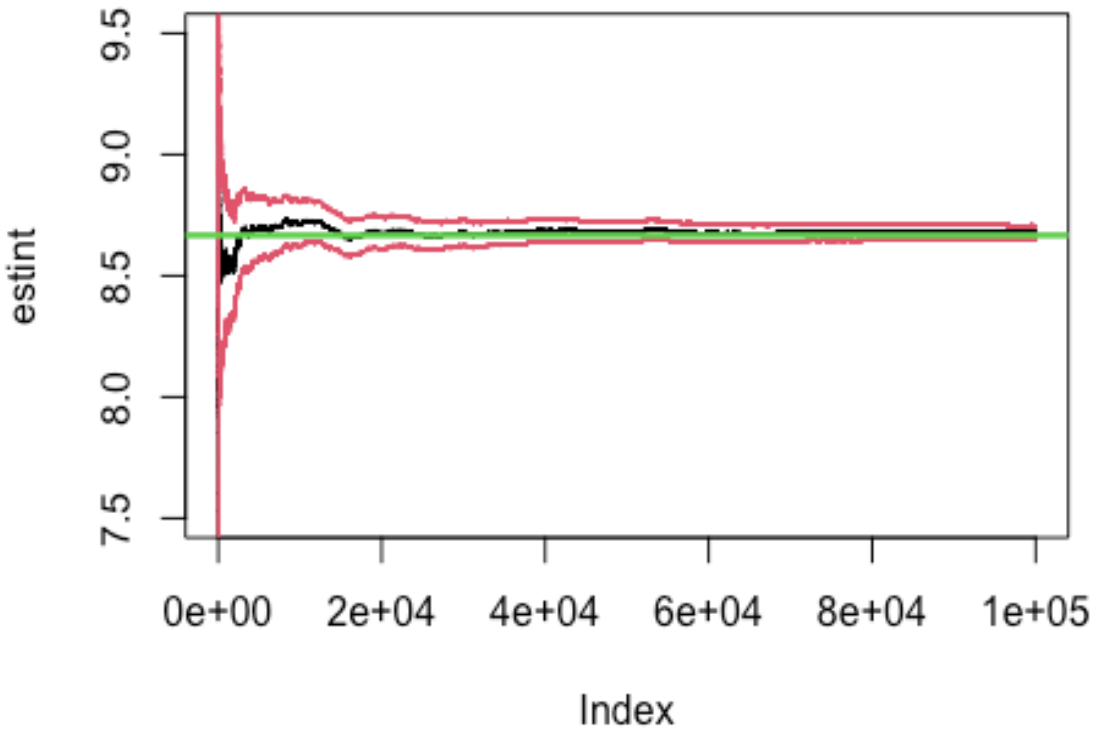
  MC.SE <- SQRT(VAR(G(X))/10^5)
  BOUNDS<-C(HAT.THETA-1.96*MC.SE,HAT.THETA+1.96*MC.SE)

  W<-G(X)*(UPLIM-LOWLIM)
  ESTINT<- CUMSUM(W)/(1:10^5) #RUNNING MEAN (CUMULATIVE ESTIMATES OF THE INT
EGRAL)
  ESTERR <- SQRT(CUMSUM((W-ESTINT)^2))/(1:10^5)
  PLOT(ESTINT,TYPE="L",YLIM=C(MINY,MAXY))
#  PLOT(ESTINT,TYPE="L",YLIM=C(MIN(ESTINT-1.96*ESTERR),MAX(ESTINT+1.96*ESTER
R)))
  LINES(ESTINT-1.96*ESTERR,COL=2)
  LINES(ESTINT+1.96*ESTERR,COL=2)

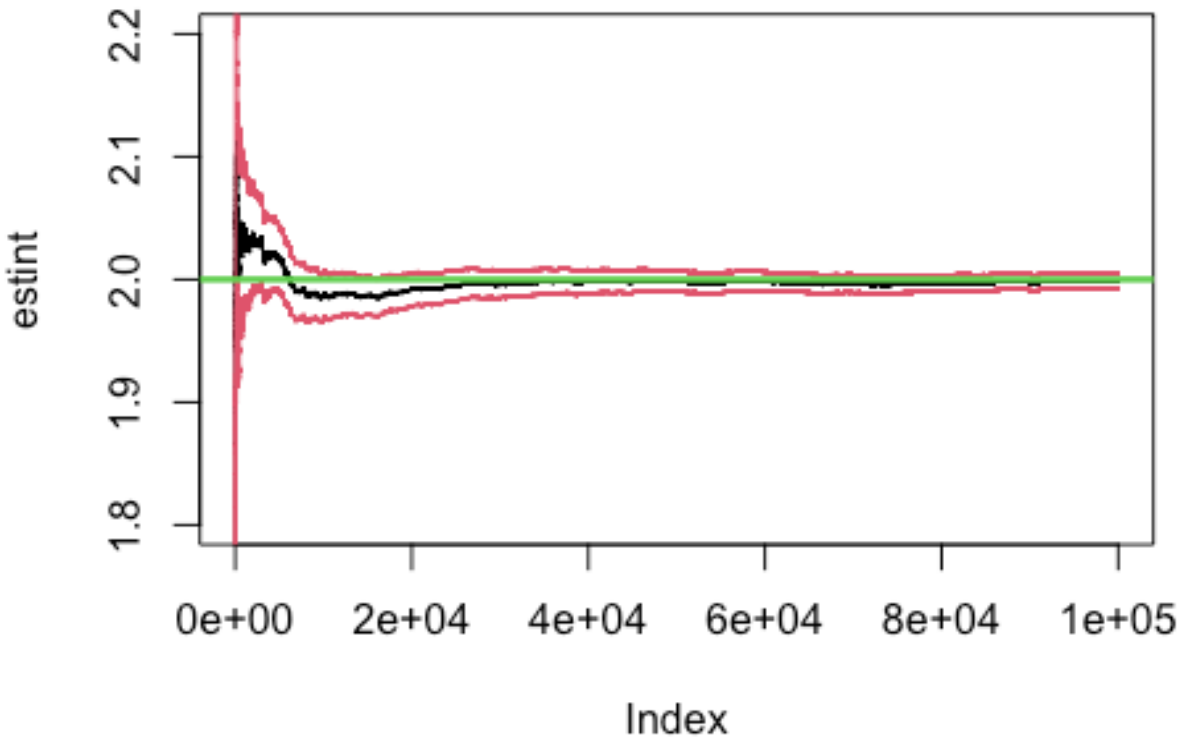
  RESULT<-LIST(MC=HAT.THETA,
               MCSE=MC.SE,
               BOUNDS=BOUNDS)

}
```

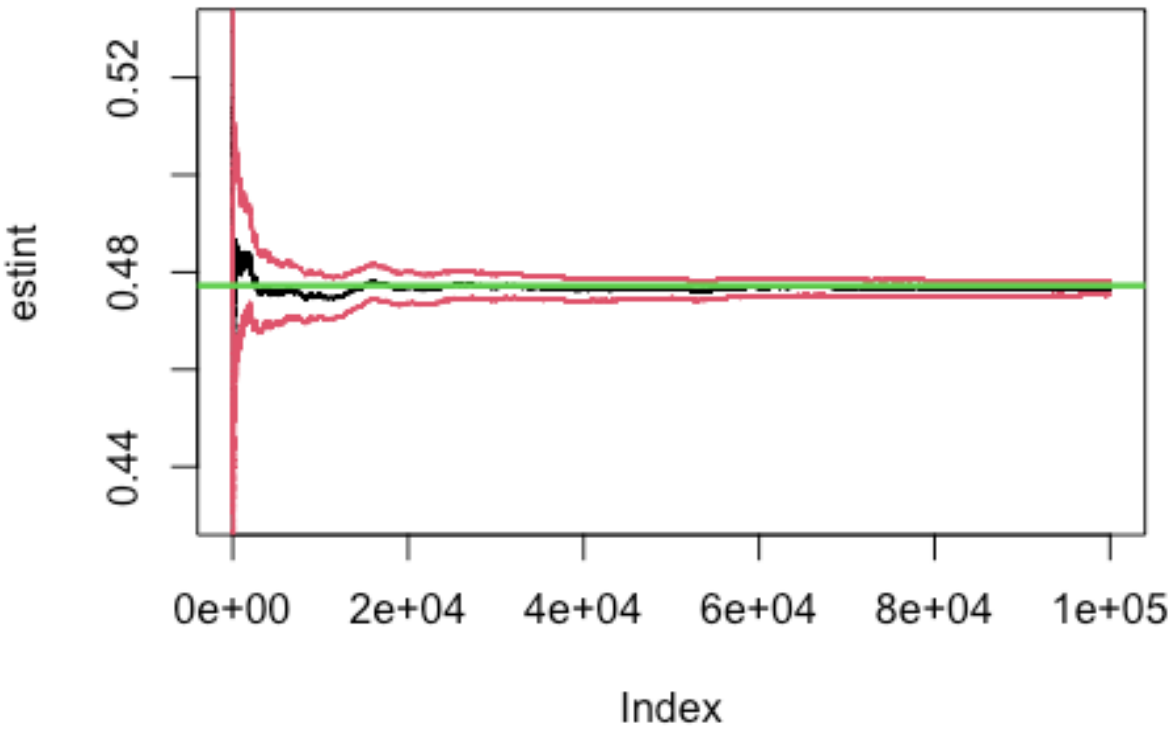
```r
MCA<- INTEGRATION1(G=FUNCTION_A, LOWLIM=1, UPLIM=3, MINY=7.5, MAXY=9.5)$MC
#THE ANTIDERIVATIVE OF X^2 IS (X^3)/3, THEN THE INTEGRAL IS EQUAL TO:
REALVA<-(3^3)/3-(1^3)/3 #8.666667
ABLINE(H=REALVA,COL=3,LWD=2.5)
```



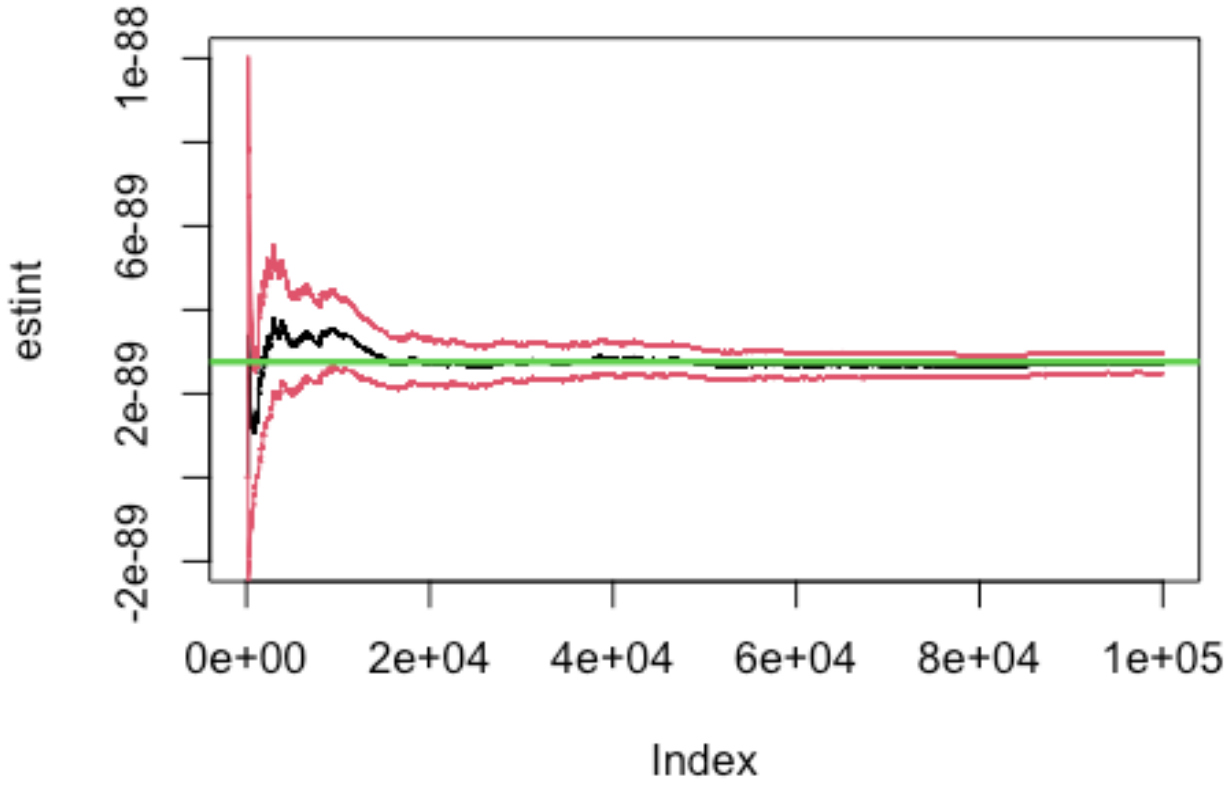

```r
MCB<- INTEGRATION1(G=FUNCTION_B, LOWLIM=0, UPLIM=PI, MINY=1.8, MAXY=2.2)$MC
#THE ANTIDERIVATIVE OF SIN(X) IS -COS(X), THEN THE INTEGRAL IS EQUAL TO:
REALVB<- (-COS(PI)+COS(0)) #2
ABLINE(H=REALVB,COL=3,LWD=2.5)
```




```
MCC<- INTEGRATION1(G=FUNCTION_C, LOWLIM=0, UPLIM=2, MINY=0.43, MAXY=0.53)$MC
#THE FUNCTION C IS THE STANDARD NORMAL PROBABILITY DENSITY FUNCTION.
#THIS INTEGRAL CALCULATES THE PROBABILITY THAT A STANDARD NORMAL VARIABLE
#IS BETWEEN 0 AND 2
REALVC<- PNORM(2,0,1)-PNORM(0,0,1)#0.47725
ABLINE(H=REALVC,COL=3,LWD=2.5)
```




```
MCD<-INTEGRATION1(G=FUNCTION_D, LOWLIM=0, UPLIM=1/20, MINY=-2E-89, MAXY=1E-8
8)$MC
#THE FUNCTION D IS THE RECIPROCAL OF THE STANDARD NORMAL DENSITY FUNCTION.
#IT DOES NOT HAVE AN ELEMENTARY CLOSED-FORM SOLUTION.
REALVD <-INTEGRATE(FUNCTION_D,0,(1/20))$VALUE #2.75E-89
ABLINE(H=REALVD,COL=3,LWD=2.5)
```




```
#MC FOR DOUBLE INTEGRALS
#MINY AND MAXY ARE GRAPHICAL PARAMETERS
INTEGRATION2 <- FUNCTION(G, LOWLIMX, UPLIMX, LOWLIMY, UPLIMY, MINY, MAXY) {
  SET.SEED(321)
  F <- FUNCTION(X,Y) DUNIF(X,LOWLIMX,UPLIMX)*DUNIF(Y,LOWLIMY,UPLIMY)
  X1 <- RUNIF(10^5, LOWLIMX, UPLIMX)
  Y1 <- RUNIF(10^5, LOWLIMY, UPLIMY)

  HAT.THETA <- MEAN(G(X1,Y1)/F(X1,Y1))

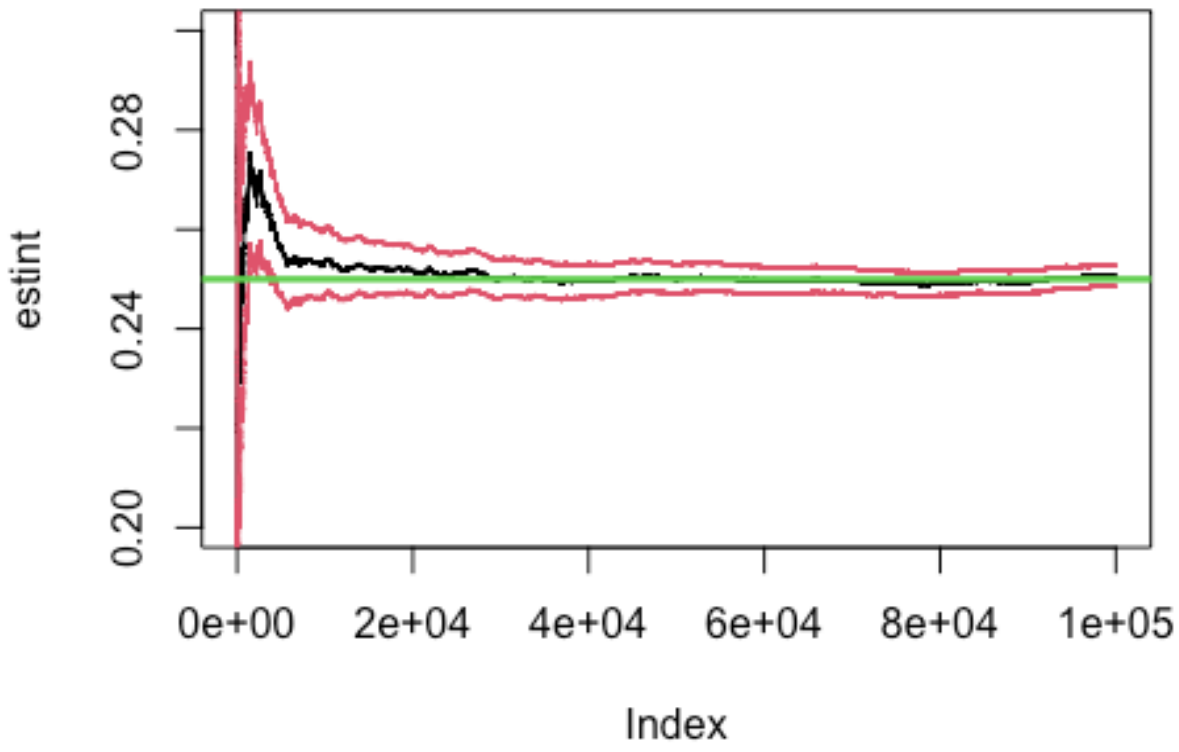

  MC.SE <- SQRT(VAR(G(X1,Y1))/10^5)
  BOUNDS<-C(HAT.THETA-1.96*MC.SE,HAT.THETA+1.96*MC.SE)

  W<-G(X1,Y1)*((UPLIMX - LOWLIMX) * (UPLIMY - LOWLIMY))
  ESTINT<- CUMSUM(W)/(1:10^5) #RUNNING MEAN (CUMULATIVE ESTIMATES OF THE INT
EGRAL)
  ESTERR <- SQRT(CUMSUM((W-ESTINT)^2))/(1:10^5)
  PLOT(ESTINT,TYPE="L",YLIM=C(MINY,MAXY))
  LINES(ESTINT-1.96*ESTERR,COL=2)
```

```
  LINES(ESTINT+1.96*ESTERR,COL=2)

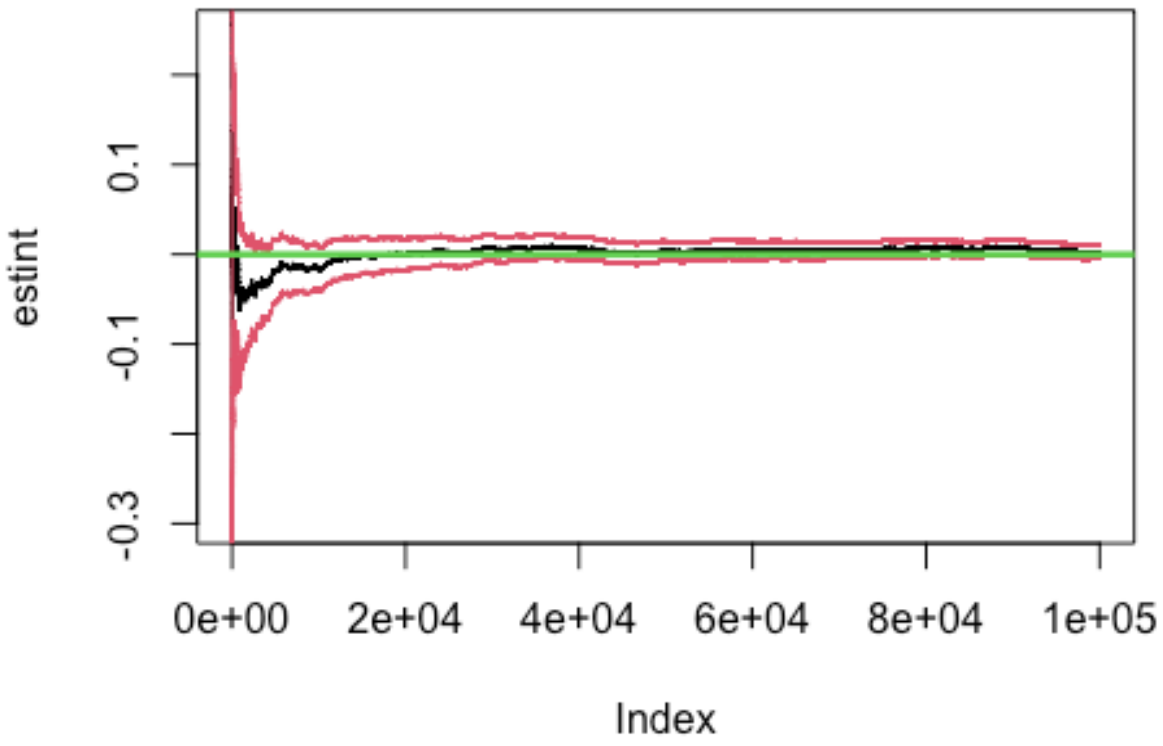
  RESULT<-LIST(MC=HAT.THETA,
               MCSE=MC.SE,
               BOUNDS=BOUNDS)
}

MCE<-INTEGRATION2(G=FUNCTION_E, LOWLIMX=0, UPLIMX=1, LOWLIMY=0, UPLIMY=1, MI
NY=0.2, MAXY=0.3)$MC
DET_INT_E <- INTEGRATE(FUNCTION(X) {
  SAPPLY(X, FUNCTION(XX) {
    INTEGRATE(FUNCTION(Y) FUNCTION_E(XX, Y), 0, 1)$VALUE
  })
}, 0, 1)
RES_E<-DET_INT_E$VALUE #0.25
ABLINE(H=RES_E,COL=3,LWD=2.5)
```

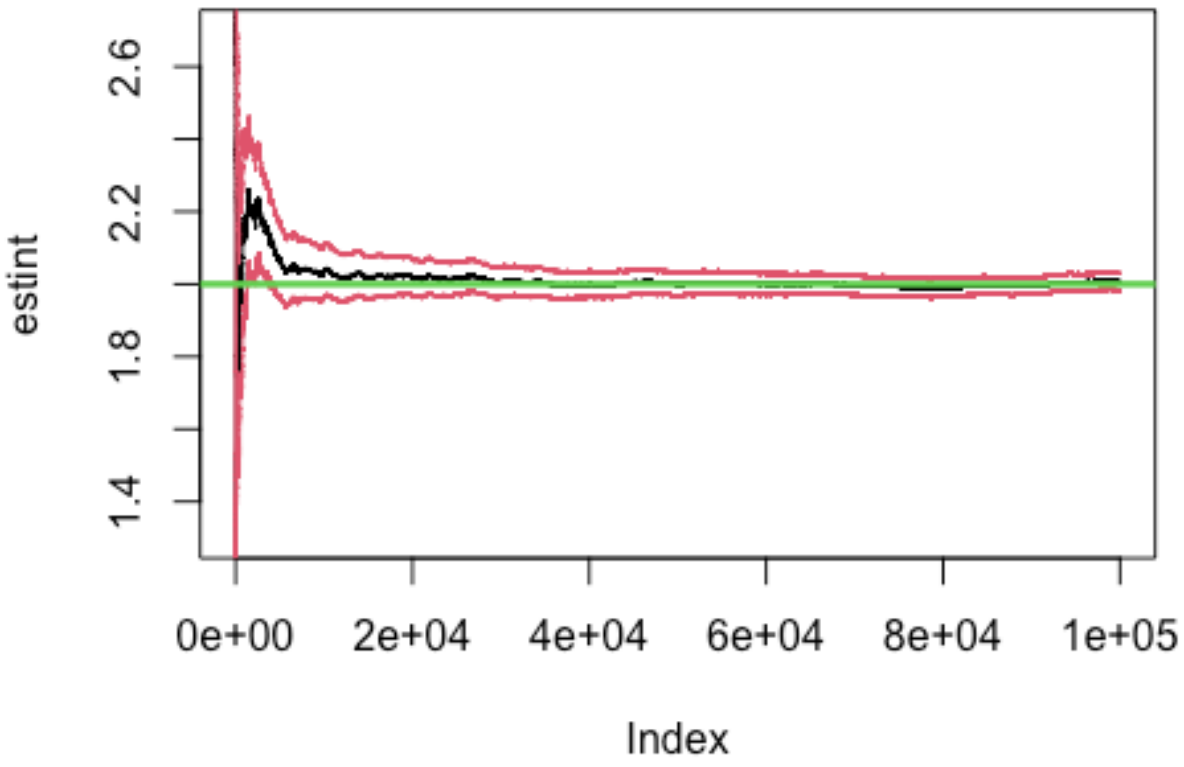



```
MCF<-INTEGRATION2(G=FUNCTION_F, LOWLIMX=0, UPLIMX=1, LOWLIMY=PI, UPLIMY=2*PI
, MINY=-0.3, MAXY=0.25)$MC
DET_INT_F <- INTEGRATE(FUNCTION(X) {
  SAPPLY(X, FUNCTION(XX) {
```

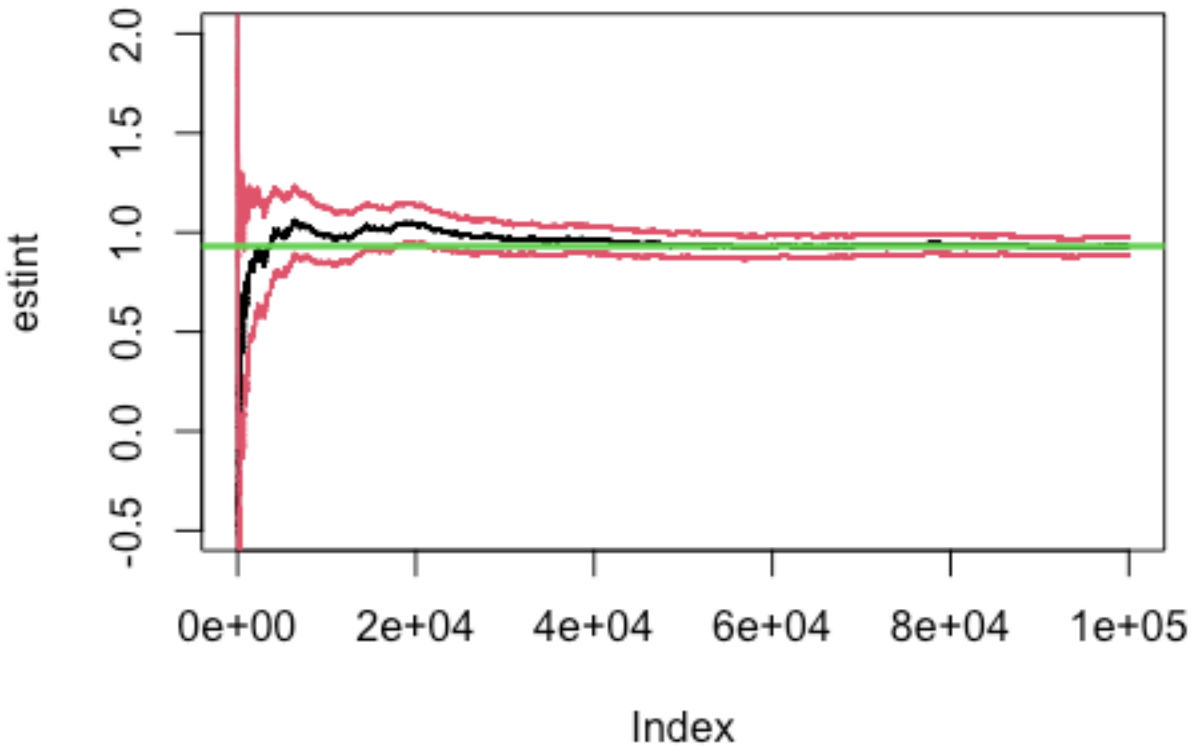
```
    INTEGRATE(FUNCTION(Y) FUNCTION_F(XX, Y), PI, 2*PI)$VALUE
  })
}, 0, 1)
RES_F<-DET_INT_F$VALUE #5.55E-17
ABLINE(H=RES_F,COL=3,LWD=2.5)
```




```
MCG<-INTEGRATION2(G=FUNCTION_G, LOWLIMX=0, UPLIMX=1, LOWLIMY=0, UPLIMY=2, MI
NY=1.3, MAXY=2.7)$MC
DET_INT_G <- INTEGRATE(FUNCTION(X) {
  SAPPLY(X, FUNCTION(XX) {
    INTEGRATE(FUNCTION(Y) FUNCTION_G(XX, Y), 0, 2)$VALUE
  })
}, 0, 1)
RES_G<-DET_INT_G$VALUE #2
ABLINE(H=RES_G,COL=3,LWD=2.5)
```




```
MCH<-INTEGRATION2(G=FUNCTION_H, LOWLIMX=0, UPLIMX=PI, LOWLIMY=-SQRT(PI), UPL
IMY=0, MINY=-0.5, MAXY=2)$MC
DET_INT_H <- INTEGRATE(FUNCTION(X) {
  SAPPLY(X, FUNCTION(XX) {
    INTEGRATE(FUNCTION(Y) FUNCTION_H(XX, Y), (-SQRT(PI)), 0)$VALUE
  })
}, 0, PI)
RES_H<-DET_INT_H$VALUE #0.9311
ABLINE(H=RES_H,COL=3,LWD=2.5)
```




```
TAB <- DATA.FRAME(
  FUNCTION = C("# SINGLE INTEGRALS","X^2", "SIN(X)", "(1 / SQRT(2*PI)) * EXP
(-X^2 / 2)",
               "EXP(-1/(2*X^2)) / (X^2 * SQRT(2*PI))", "# DOUBLE INTEGRALS",
 "X*Y*(X^2 + Y^2)",
               "X*SIN(X*Y)", "X^2*Y*EXP(X*Y)", "X*COS(X^2 + Y)"),
  "MONTE CARLO" = C("-", ROUND(MCA,5), ROUND(MCB,5), ROUND(MCC,5), ROUND(MCD
,5), "-", ROUND(MCE,5), ROUND(MCF,5), ROUND(MCG,5), ROUND(MCH,5)),
  "REAL VALUE"= C("-", ROUND(REALVA,5), ROUND(REALVB,5), ROUND(REALVC,5), RO
UND(REALVD,5),
                                      "-", ROUND(RES_E,5), ROUND(RES_F,5), ROUN
D(RES_G,5), ROUND(RES_H,5)),
  "DIFFERENCE"= C("-", ROUND(MCA-REALVA, 5), ROUND(MCB-REALVB, 5), ROUND(MCC
-REALVC,5), ROUND(MCD-REALVD, 5),
                  "-", ROUND(MCE-RES_E,5), ROUND(MCF-RES_F, 5), ROUND(MCG-RE
S_G, 5), ROUND(MCH-RES_H,5)))
TAB

##                               FUNCTION MONTE.CARLO REAL.VALUE DIFFERENCE
## 1                  # SINGLE INTEGRALS           -          -          -
## 2                                  X^2     8.67735    8.66667    0.01068
```

```
## 3                                  SIN(X)     1.99894          2   -0.00106
## 4       (1 / SQRT(2*PI)) * EXP(-X^2 / 2)     0.47673    0.47725   -0.00052
## 5  EXP(-1/(2*X^2)) / (X^2 * SQRT(2*PI))           0          0          0
## 6                     # DOUBLE INTEGRALS           -          -          -
## 7                        X*Y*(X^2 + Y^2)     0.25055       0.25    0.00055
## 8                             X*SIN(X*Y)     0.00316          0    0.00316
## 9                          X^2*Y*EXP(X*Y)     2.00634          2    0.00634
## 10                         X*COS(X^2 + Y)     0.92997     0.9311   -0.00113
```

Overall, the Monte Carlo (MC) estimates are in close agreement with the true or deterministic values.

The difference is generally very small across all integrals. The Monte Carlo method demonstrates good convergence for all tested integrals, as evidenced by the convergence graphs The convergence to the true value was monitored with 10^5 samples and 95% confidence bounds.

For all eight functions the deterministic or real value (represented by the green line convergence plots) successfully falls within the 95% confidence interval (delineated by the red lines).

4. *Use the inverse transformation method to generate values from a X~Weibull distribution with:*

$$F(x) = 1-\exp(-x\alpha/\beta),\ x\geq 0$$

with α = 1 and β = 5. Graph the histogram of the sample with the theoretical density superimposed.

Use the rejection algorithm to generate nrandom observations from X using the Cauchy distribution as proposal. Find and adequate value of the location l and the scale s parameters of the Cauchy distribution so as to obtain a good acceptance rate.

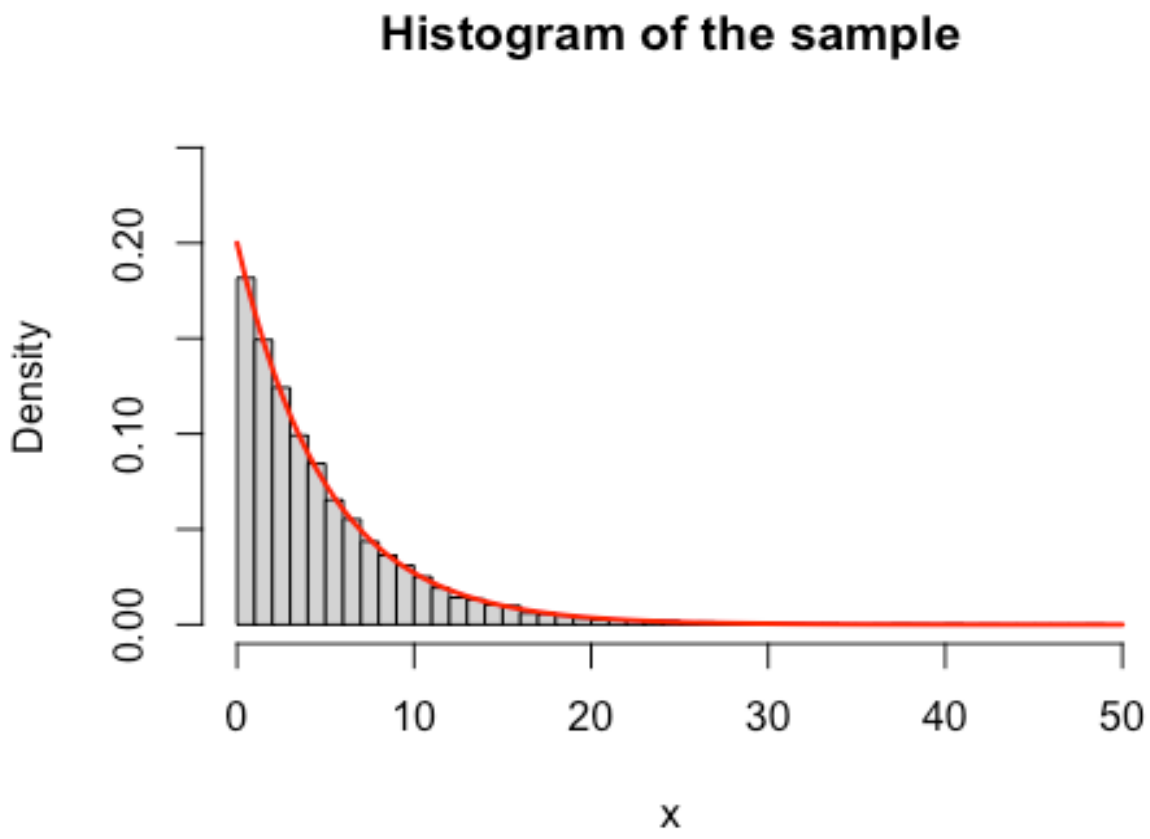
```r
SET.SEED(123)
ALPHA <- 1
BETA <- 5

# GENERATE USING THE INVERSE TRANSFORMATION METHOD
U <- RUNIF(N)

# INVERSE CDF
X_INV <- (-BETA * LOG(1-U))^(1/ALPHA)

# THEORETICAL DENSITY FUNCTION
WEIBULL <- FUNCTION(X, ALPHA, BETA) {
  (ALPHA/BETA) * X^(ALPHA - 1) * EXP(-X^ALPHA/BETA)
}
#WE DON'T USE DWEIBULL() BECAUSE IT USES DIFFERENT PARAMETERIZATION
```

```r
#HISTOGRAM
HIST(X_INV, BREAKS = 50, PROBABILITY = TRUE,
     MAIN = "HISTOGRAM OF THE SAMPLE",
     XLAB = "X", YLIM = C(0, 0.25))
CURVE(WEIBULL(X, ALPHA, BETA), FROM=0, TO=50, COL="RED", LWD=2, ADD=T)
```




```r
### REJECTION SAMPLING ALGORITHM
# TARGET: WEIBULL(A=1, B=5)
# PROPOSAL: CAUCHY(L=5, S=5)

SET.SEED(1234)

# PARAMETERS
U <- RUNIF(N)
L <- 4
S <- 4

#FUNCTIONS
F <- FUNCTION(X) IFELSE(X >= 0, (ALPHA / BETA) * X^(ALPHA - 1) * EXP(-X^ALPH
A / BETA), 0)
G <- FUNCTION(X) DCAUCHY(X, LOCATION = L, SCALE = S)
```

```r
#CONSTANT C
RATIO <- FUNCTION(X) {F(X) / G(X)}
OPT <- OPTIMIZE(RATIO, INTERVAL = C(0, 40), MAXIMUM = TRUE)
C <- OPT$OBJECTIVE
C

## [1] 1.139995

# REJECTION SAMPLING ALGORITHM
SET.SEED(1234)

REJECTSAMP <- FUNCTION(N, C=C, F=F, G=G){
  X <- NUMERIC(N); K <- 0; J <- 0
  WHILE (K < N){
    J <- J + 1
    Y <- RCAUCHY(1, LOCATION = L, SCALE = S)
    U <- RUNIF(1)
    IF (U <= F(Y) / (C * G(Y))) {
      K <- K + 1
      X[K] <- Y
    }
  }
  OUT <- LIST(X = X, J = J)
  RETURN(OUT)
}

RJ <- REJECTSAMP(N, C=6.282984, F=F, G=G)

# ACCEPTANCE RATE
N/RJ$J

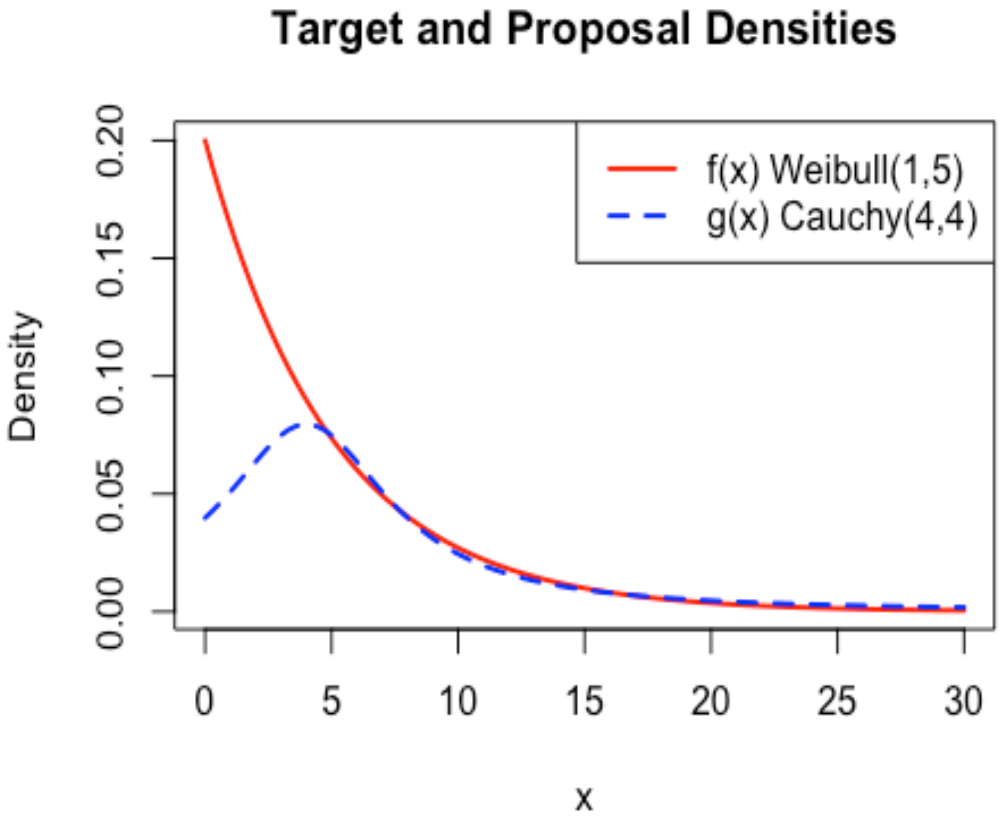
## [1] 0.1605523
```

In the target distribution Weibull(α = 1, β = 5), the density is highest at x = 0, but most of the mass is concentrated around values between 4 and 5. If the peak of the proposal distribution (l) is too far to the right, many generated values fall in regions where the target density is low, resulting in numerous rejections and a low acceptance rate. After some trials, we decided to set l = 4 and s = 4, to center the peak of the Cauchy distribution near the region where the Weibull has most of its mass, while ensuring enough width to cover the tail.

This choice represents a good compromise between coverage of the target density and algorithmic efficiency, allowing for a higher acceptance rate without compromising sampling correctness.

```r
#PLOT
CURVE(F(X), FROM=0, TO=30, COL="RED", LWD=2,
      YLAB="DENSITY", XLAB="X", MAIN="TARGET AND PROPOSAL DENSITIES")
CURVE(G(X), FROM=0, TO=30, COL="BLUE", LWD=2, LTY=2, ADD=TRUE)

LEGEND("TOPRIGHT", LEGEND=C("F(X) WEIBULL(1,5)", "G(X) CAUCHY(4,4)"),
       COL=C("RED","BLUE"), LWD=2, LTY=C(1,2))
```


Target and Proposal Densities
f(x) Weibull(1,5)
g(x) Cauchy(4,4)
Density
0.00
0.05
0.10
0.15
0.20
0
5
10
15
20
25
30
x